

1							
2							
3							
4							
5							
6							
7							
8							
9							
0	☒						

Registration number

--

Signature

Note:

- Cross your Registration number(with leading zero). It will be evaluated automatically.
- Sign in the corresponding signature field.

Principles of Computer Vision

Exam: IN2133 / Endterm

Date: Thursday 27th February, 2025

Examiner: Prof. Dr.-Ing. Darius Burschka

Time: 11:00 – 12:00

Working instructions

- This exam consists of **12 pages** with a total of **2 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 135 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **non-programmable pocket calculator**
 - one **analog dictionary** English ↔ native language
- Subproblems marked by * can be solved without results of previous subproblems.
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

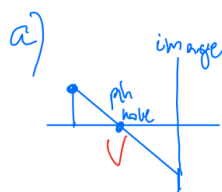
Left room from _____ to _____ / Early submission at _____

Problem 1 Image Formation (78 credits)

A camera with the perspective camera model is used to acquire images of the environment. A lens with the focal length $f_L = 8\text{mm}$ is used in front of a 512×512 pixels CCD chip with a pixel-size $p_x = 7\mu\text{m}$.

a)* Lens system

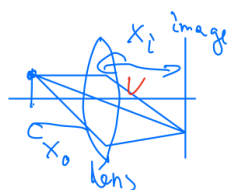
- Sketch briefly the difference in the collected light between a primitive pinhole camera and a camera with a lens mounted in front of the CCD chip;
- List in a bulleted list, what is improved and what additional problems get caused by adding a lens to the system (name 1 positive and 3 negative effects);
- Explain with additional drawing, what causes a limited distance range in front of the camera that creates a sharp image and name this effect;
- How can this range be extended? Add to the drawing a solution that will extend this distance range and provide the correct name for this solution.



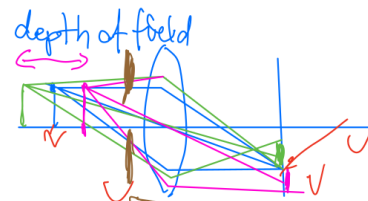
single ray ✓
illuminates

pro
- more light (a cone) ✓
illuminates
the image

con
- only a specific $\frac{1}{x_o} + \frac{1}{x_i} = \frac{1}{f}$ ✓
results in a sharp image
- quantization of CCD results in
limited depth of field ✓
- lens distortions, aberration ✓
focal point not in center ✓



a cone of light
illuminates ✓



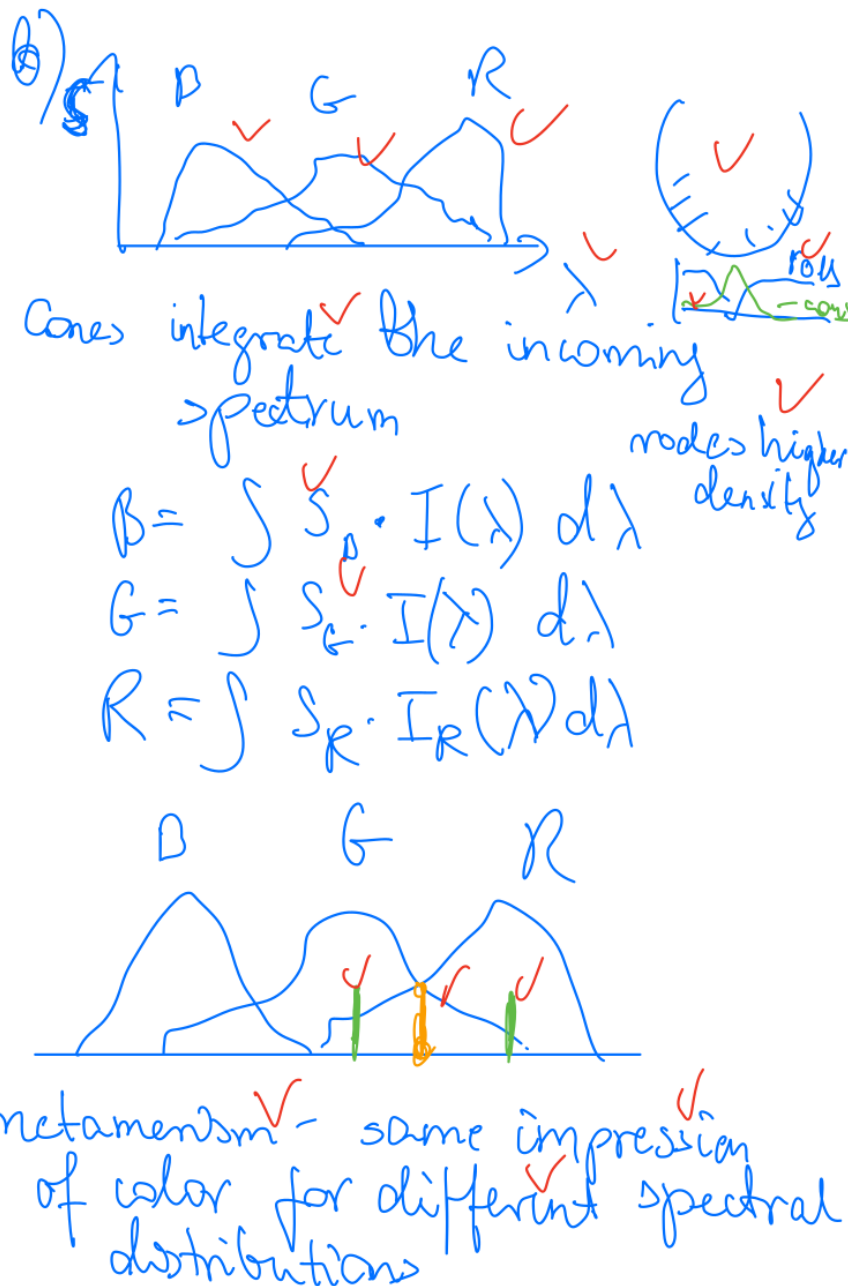
objects not in optimal distance
to lens ✓ smear. if smear
smaller than pixel size then still
sharp

Extension through iris limiting
the opening of the cone (more
light required)

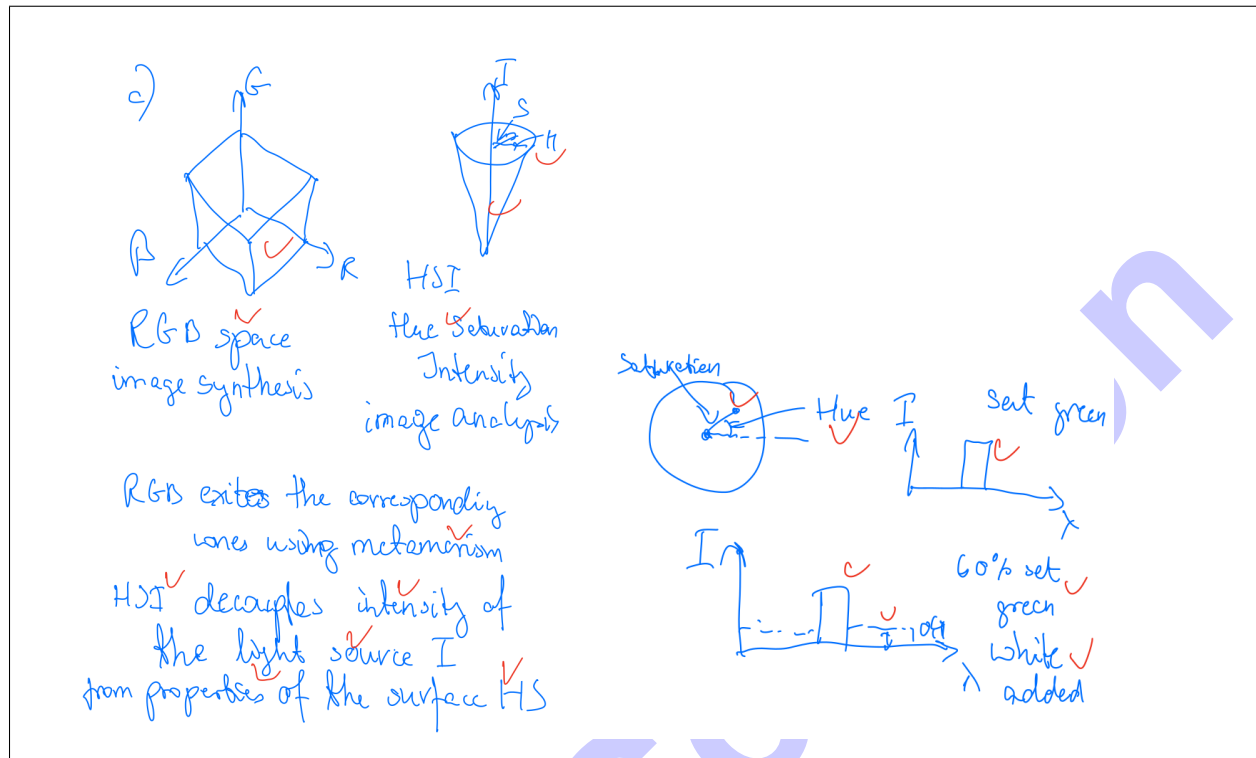
b)* A human eye senses the incoming light using various photosensitive elements on the retina. The eye is capable of sensing a *visible spectrum* using this structure.

- Explain and draw, how the eye manages to sense the *visible spectrum* with a limited number of sensing elements in each sensed pixel naming the corresponding elements
- Sketch the spectral sensitivity of elements responsible for color perception
- Explain "metamerism" sketching examples of contributing spectral distributions of the incoming light.

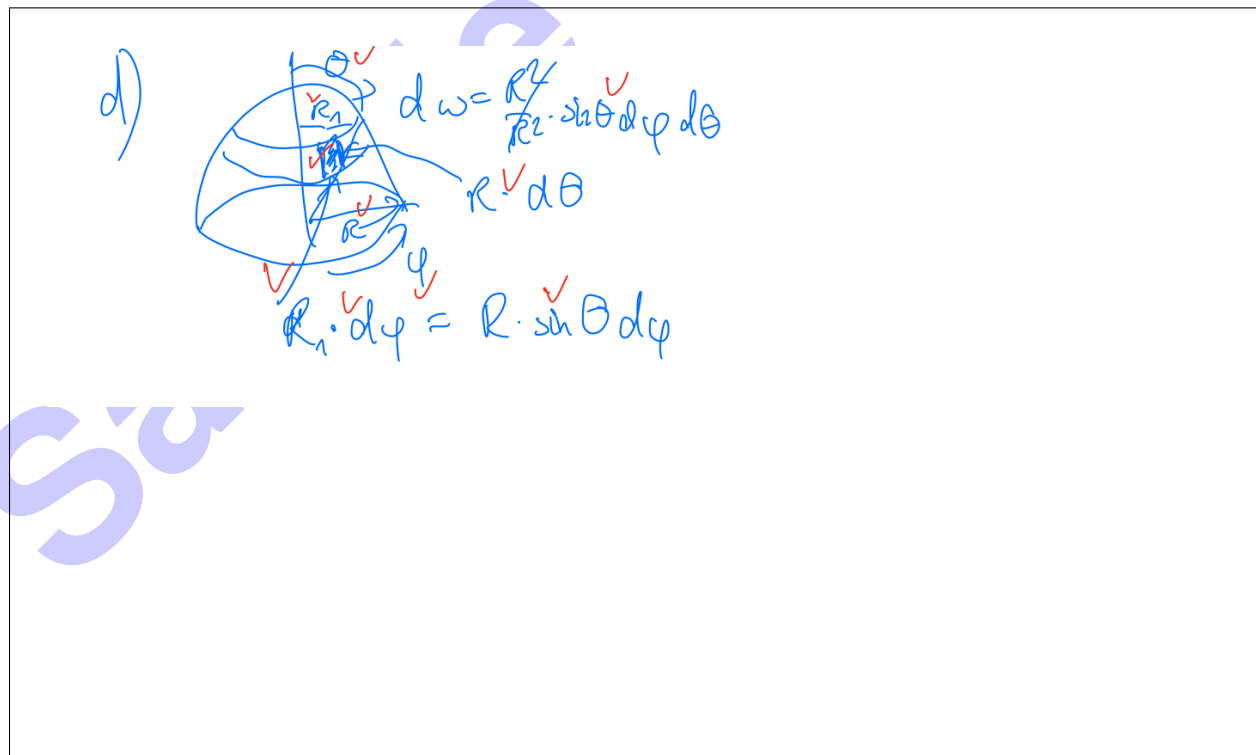
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17



c)* Draw the color spaces used for image synthesis on screens and for image analysis and explain, why not the same color space is used? What is the advantage of the color space used for analysis? Explain the term *saturation* and *hue* in a separate drawing and draw spectral distributions for fully saturated green and 60% saturated green.



d)* How can you express the solid angle $d\omega$ that is used to integrate the incoming light energy on a patch using the angles of the hemisphere. Draw the parameterization of the solid angle on the hemisphere and derive the equation for $d\omega$:



e)* Consider a rectangular uniformly shining surface with diagonal corners at $(w, d/2, 0)^T, (w, d/2, h)^T, (w, -d/2, h)^T, (w, -d/2, 0)^T$ as depicted in Figure below. The total radiance of the entire rectangle is I .

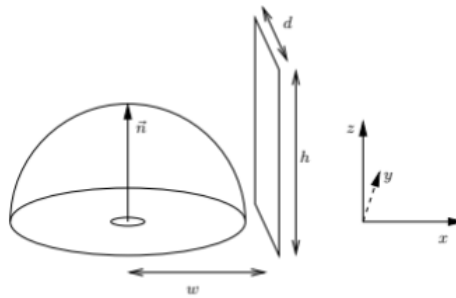


Figure 1: Problem setup.

The intensity E illuminating a point in the middle of the half-sphere H_z can be calculated as

$$E = \int_{D \in H_z} L(\omega) \cos \theta d\omega$$

- Which effect is reflected in the cosine expression estimating E above?
- Explain, how the expression $d\omega$ can be calculated for a half-sphere with the radius $r=1$ using the azimuth angle ϕ and elevation angle Θ ?
- What is the intensity arriving at the center of the half-sphere $(0,0,0)$ with normal vector $(0,0,1)$ (see Figure) from a point $p = (x, y, w)^T$ on the rectangle? Try to express the function in the integral above a function of (w,y,z,dz,dy) .
- How does the full integral including integration boundaries look for this example?

e) cosine expression ✓
explains the foreshortening effect ✓

$$\begin{aligned} E &= \int_{\frac{-d}{2}}^{\frac{d}{2}} \int_0^h L \cos \theta d\omega = \\ &= \int_{\frac{-d}{2}}^{\frac{d}{2}} \int_0^h L \cdot \frac{w}{\sqrt{w^2 + y^2 + z^2}} d\omega dz dy \\ &= \int_{\frac{-d}{2}}^{\frac{d}{2}} \int_0^h \frac{L}{\sqrt{w^2 + y^2 + z^2}} \cdot \frac{2}{\sqrt{w^2 + y^2 + z^2}} dz dy \end{aligned}$$

$$d\omega = \sin \theta d\phi d\theta$$

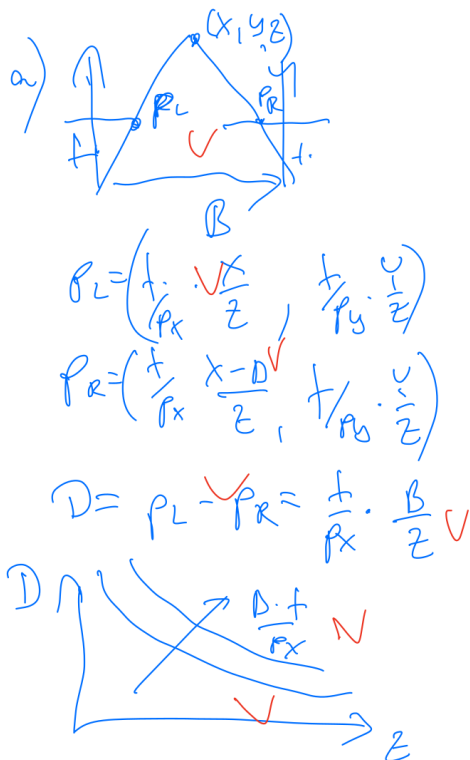
Sample Solution

Problem 2 Reconstruction (57 credits)

a)* A binocular camera system is used to measure depth of objects in range $z = [1 - 3]$ with an accuracy better than $\Delta z = 0.01m$. The cameras have both identical focal lengths $f/p_x = 800pixels$ and the matching points in the images are detected with an accuracy of 0.5pixels

- How can you calculate z (along optical axis) for an ideal binocular system from matched observations?
- Draw the curve relating the z -distance of the reconstructed points to the value estimated from the images.
- How can you derive the depth variation relative to image signal variation of the binocular system?
- Based on above equation, what is the baseline B of the system that needs to be chosen to get the required accuracy?

☐	0
☐	1
☐	2
☐	3
☐	4
☐	5
☐	6
☐	7
☐	8
☐	9
☐	10
☐	11
☐	12
☐	13



$$\frac{dd}{dz} = - \frac{B \cdot f}{p_x \cdot z^2}$$

worst $z = 1$

$$\Delta d = \left| \frac{dd}{dz} \right| \cdot \Delta z$$

$$0.5 = \left| \frac{B \cdot 800}{3^2} \right| \cdot 0.01$$

$$\frac{0.5 \cdot 3^2}{800} = B = \frac{450}{800} = 56.25$$

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11

b)* Explain using equations, how the inner structure of the Essential Matrix can be derived. What does this matrix relate to each other?

b)

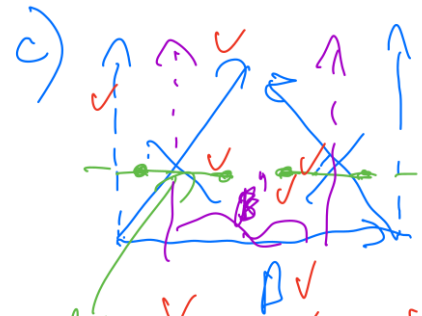


$$\begin{aligned} \vec{N} &= \vec{T} \times \vec{P_R} \\ &= \text{sk}(\vec{T}) \cdot \vec{P_R} \\ \vec{N} &= R \cdot \text{sk}(\vec{T}) \cdot \vec{P_R} \\ \vec{P_L}^T \cdot \vec{N} &= 0 = \vec{P_L}^T \cdot R \cdot \text{sk}(\vec{T}) \cdot \vec{P_R} \\ &\quad \underbrace{\vec{P_L}^T \cdot R}_{E} \end{aligned}$$

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10

c)* Show, why verged binocular camera system increases the spatial resolution of the stereo system compared to a co-planar system setup with similar field of view. Use the sensitivity equation dd/dz discussed in the lecture.

c)



rectified image

rectified system has larger Baseline B than B1

d)*

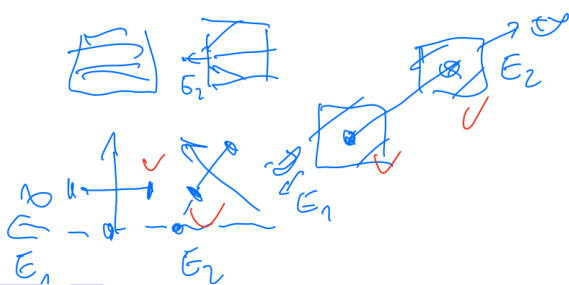
- What are epipolar lines and how are they defined?
- Sketch the camera arrangement (position and orientation of image planes, focal points and optical axes) for the following 2 epipolar geometries.
- Justify your choice by explaining where the epipoles lie and how they can be found.

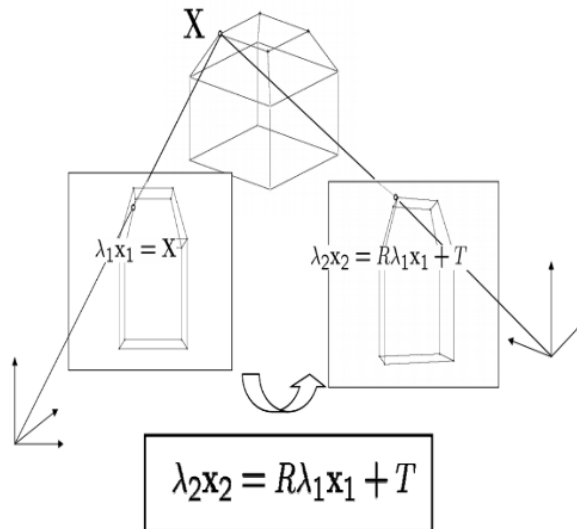


d)

epipolar line
linear search space
for corresponding points

Start at epipole
defined by intersection
of translation vector T
with image plane





e)* A point observed in two images

- Derive the equation that estimates the distances λ_1, λ_2 along the rays of observation in both cameras.
- Explain, while a pseudo-inverse is used in the solution and what does it mean for non-ideal calibration of the camera system.

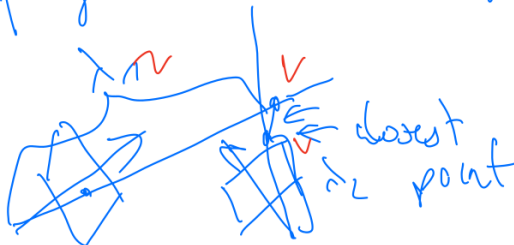
$$e) \lambda_2 \vec{n}_2 = \tilde{R} \lambda_1 \vec{n}_1 + \vec{T} \quad \checkmark$$

$$\lambda_2 \vec{n}_2 - \tilde{R} \lambda_1 \vec{n}_1 = \vec{T} \quad \checkmark$$

$$\begin{pmatrix} \vec{n}_2 & -\tilde{R} \vec{n}_1 \end{pmatrix} \begin{pmatrix} \lambda_2 \\ \lambda_1 \end{pmatrix} = \vec{T}$$

$$\begin{pmatrix} \lambda_2 \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} \vec{n}_2 & -\tilde{R} \vec{n}_1 \end{pmatrix}^{-1} \vec{T}$$

pseudo-inverse $\checkmark$ because
over-determined system
of equations $\checkmark$ 2 unknowns 3 eq.



Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

A large rectangular area filled with a fine grid of squares, intended for writing solutions. A large, light blue, semi-transparent watermark with the text "Sample Solution" is oriented diagonally from the bottom-left to the top-right across the entire grid.

Sample Solution